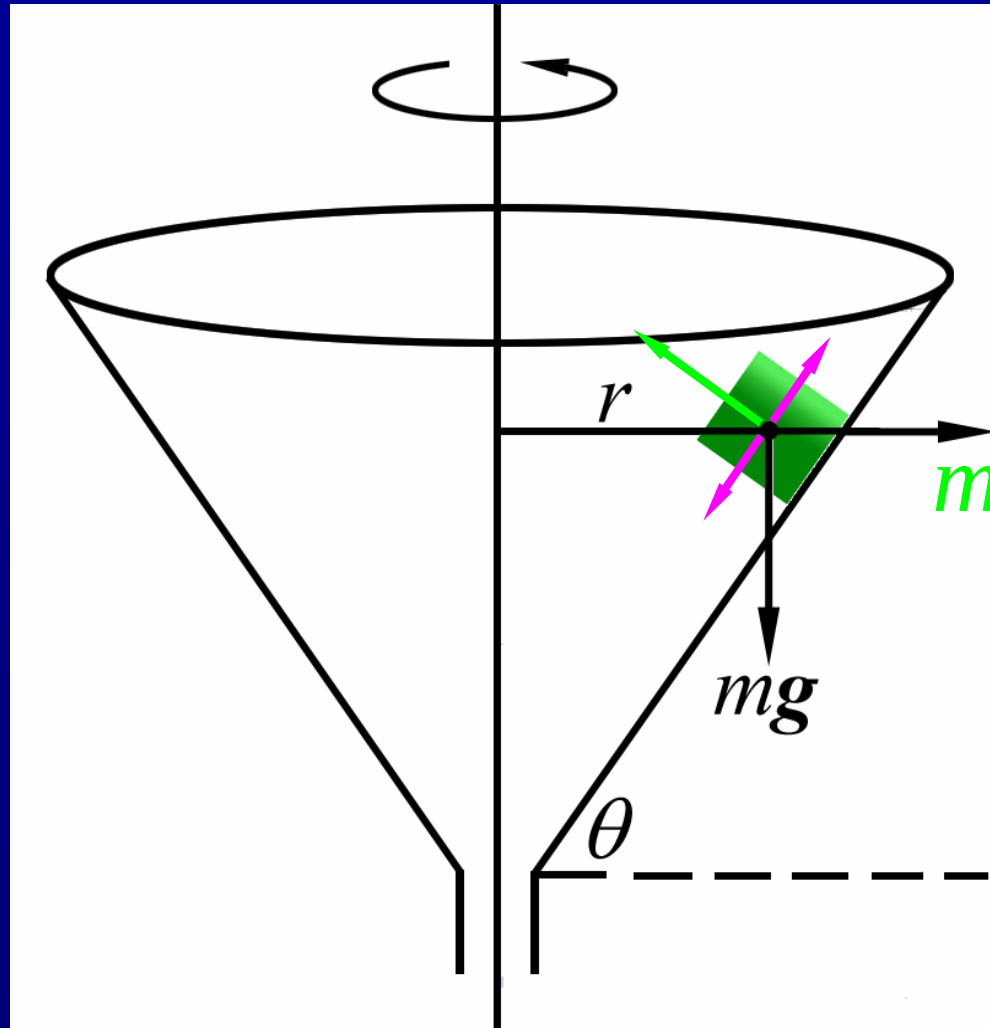


Problem 3.4 Rotating funnel

$$2\pi\nu$$



$$m(2\pi\nu)^2 r$$

$$\nu_1 \leq \nu \leq \nu_2$$

not sliding out

$$F_N - m(2\pi\nu)^2 r \sin \theta - mg \cos \theta = 0$$

$$m(2\pi\nu)^2 r \cos \theta - mg \sin \theta - F_f = 0$$

$$F_f \leq \mu_s F_N$$

$$\cancel{m(2\pi\nu)^2 r \cos \theta} - \cancel{mg \sin \theta} \leq \mu [\cancel{m(2\pi\nu)^2 r \sin \theta} + \cancel{mg \cos \theta}]$$

$$(2\pi\nu)^2 r (\cos \theta - \mu \sin \theta) \leq g (\sin \theta + \mu \cos \theta)$$

Condition of not sliding out is:

$$(2\pi\nu)^2 \leq \frac{g}{r} \frac{\sin \theta + \mu \cos \theta}{\cos \theta - \mu \sin \theta} \equiv (2\pi\nu_2)^2$$

$$(2\pi\nu)^2 \leq \frac{g \sin \theta + \mu \cos \theta}{r \cos \theta - \mu \sin \theta} \equiv (2\pi\nu_2)^2$$

$$\cos \theta - \mu \sin \theta \geq 0 \quad \text{or} \quad \cot \theta \geq \mu$$

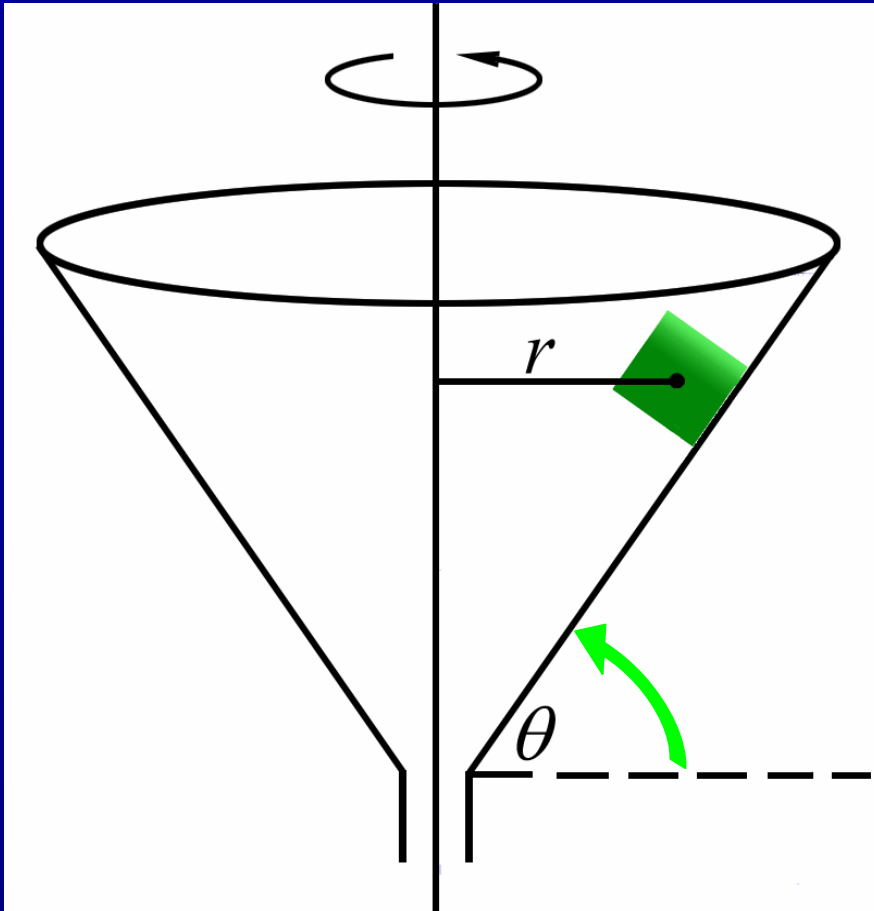
Conditional!

$$\mu > \cot \theta \quad ?$$

$$(2\pi\nu)^2 r (\cos \theta - \mu \sin \theta) \leq g (\sin \theta + \mu \cos \theta)$$

Never slides out

$$\theta < 0 \quad ?$$



If $\sin \theta + \mu \cos \theta > 0$

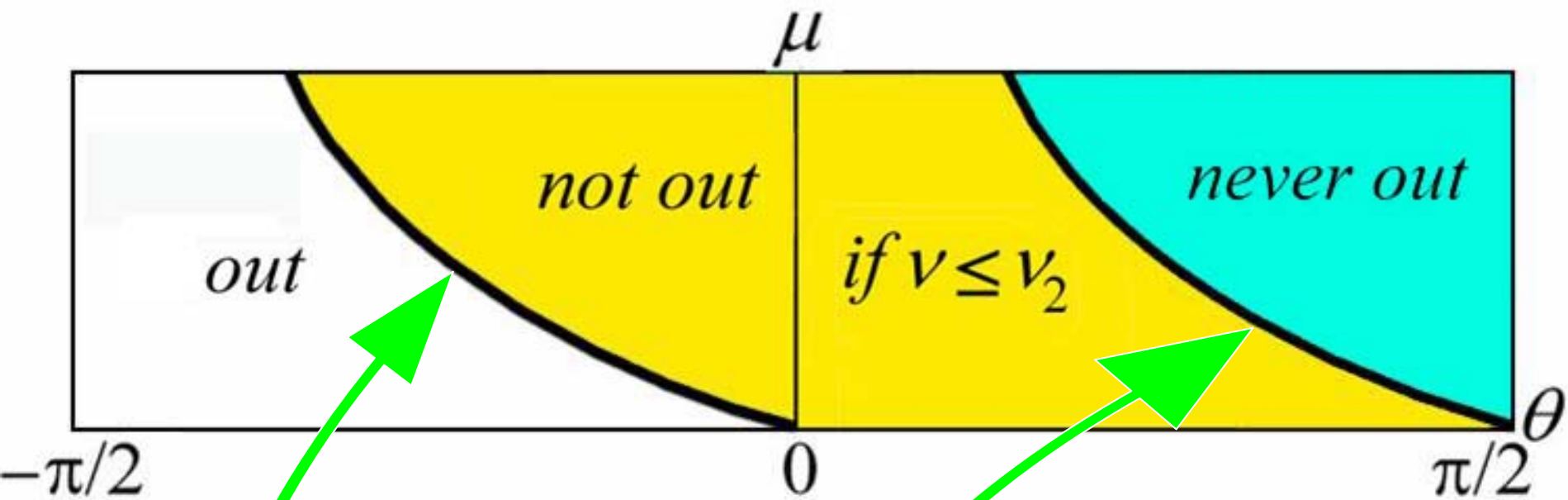
or

$-\sin(-\theta) + \mu \cos \theta > 0$

inequality leads to

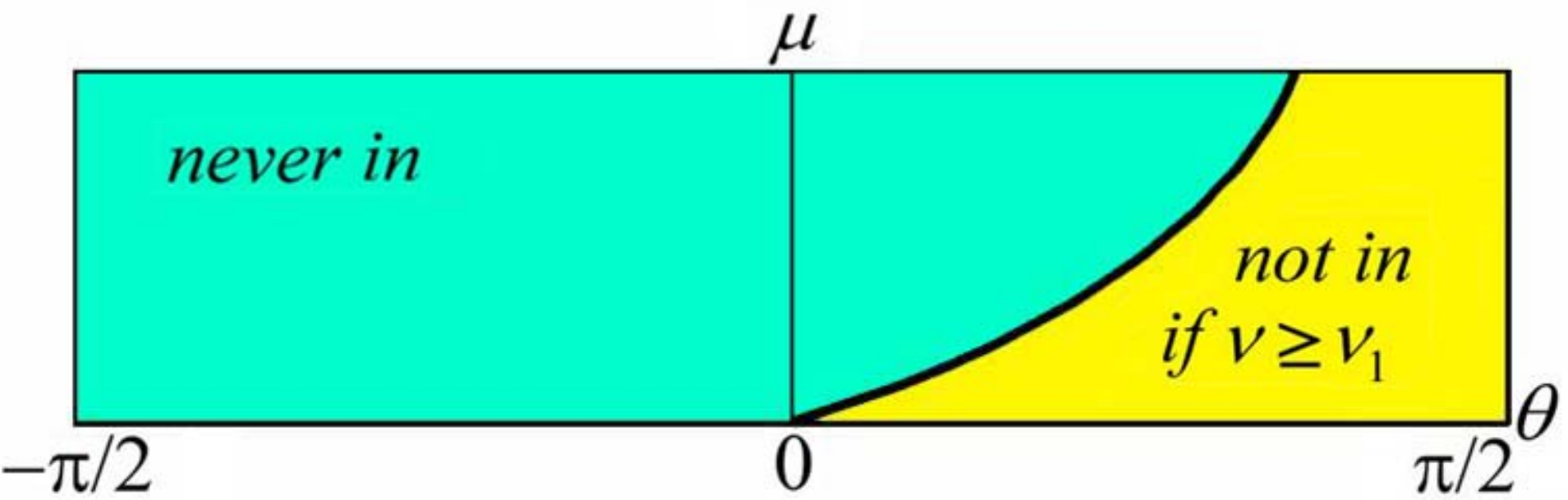
$$v < v_2$$

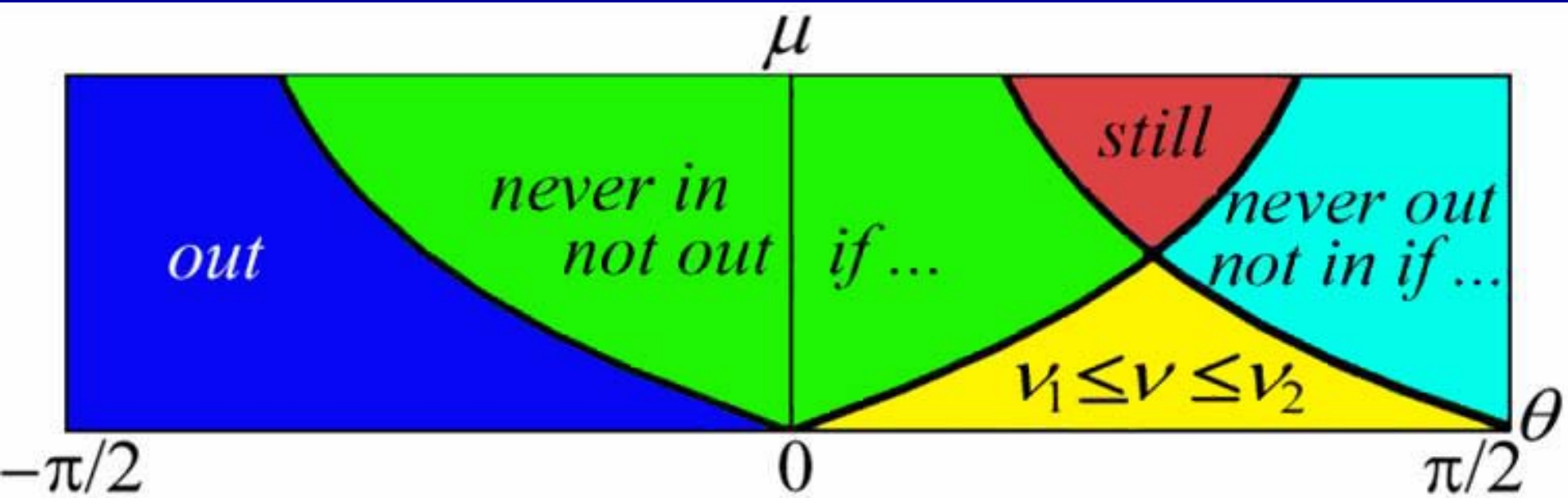
$$\mu > \frac{\sin(-\theta)}{\cos(\theta)}$$



$\mu = -\tan \theta$

$\mu = \cot \theta$





Problem 3.5

$$mg - k \frac{dz}{dt} = m \frac{d^2 z}{dt^2}$$

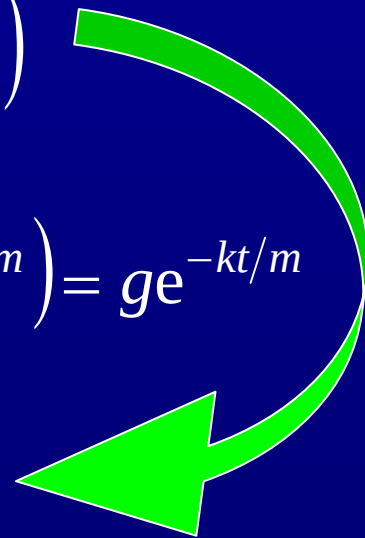
a) $\left. \frac{dz}{dt} \right|_0 = 0$; $\frac{d^2 z}{dt^2} > 0$ until $mg = k \frac{dz}{dt}$

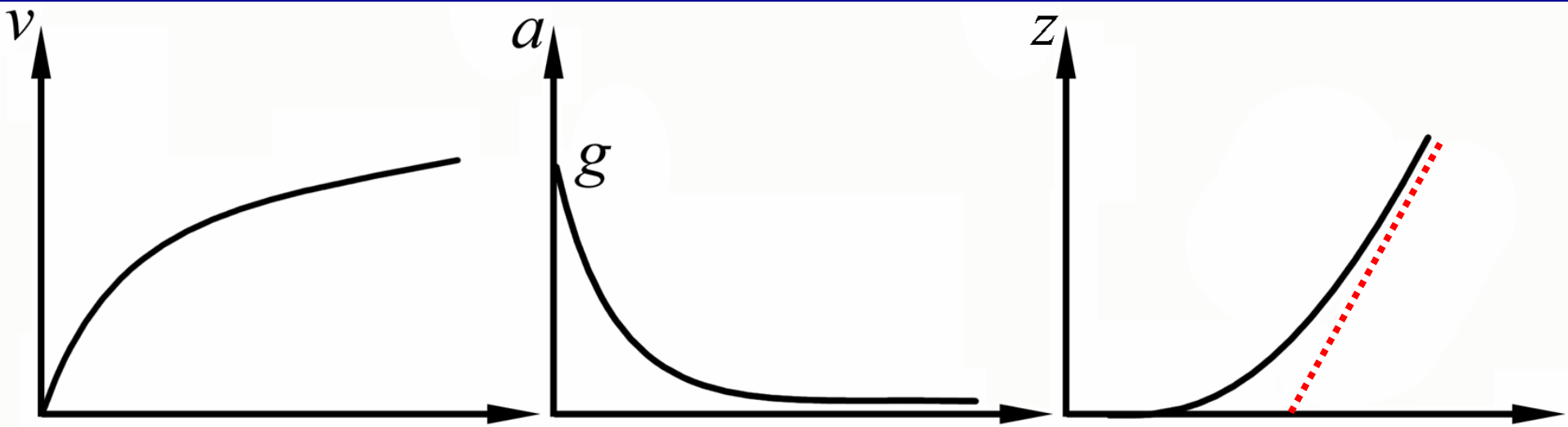
The terminal speed is determined by $v_T = \frac{mg}{k}$

b) $\frac{dv}{v_T - v} = \frac{k}{m} dt \quad v = v_T (1 - e^{-kt/m})$

c) $a = g - \frac{k}{m} v = g - \frac{k}{m} v_T (1 - e^{-kt/m}) = g e^{-kt/m}$

d) $z = v_T t + v_T \left(\frac{m}{k} \right) (e^{-kt/m} - 1)$





Asymptotically

$$z = v_T t + v_T \left(\frac{m}{k} \right) \left(e^{-kt/m} - 1 \right)$$

$$\xrightarrow{t \rightarrow \infty} v_T t - v_T \left(\frac{m}{k} \right) = v_T \left(t - \frac{m}{k} \right)$$

problem 3.9 *Only Coriolis force is considered:
(x-east; y-north; z-up)

$$-2m\omega \times \mathbf{v} = -2m\omega v_z \sin \theta \mathbf{i} \quad v_z \approx -gt$$

Equation of motion is $m\ddot{x} = 2m\omega gt \sin \theta$

Solution is $x = \frac{1}{3}\omega gt^3 \sin \theta + 0 + 0$

Substitution of $t = \sqrt{2h/g}$ **yields**

$$x = \frac{2\sqrt{2}}{3} \frac{\omega}{\sqrt{g}} \sin \theta h^{3/2} = 2.20 \times 10^{-5} h^{3/2} \sin \theta$$

problem3.10

$$F_{\text{on water}} \Delta t = -\mu u \Delta t - \mu u \Delta t$$

$$F_{\text{on blade}} = -F_{\text{on water}} = 2\mu u$$

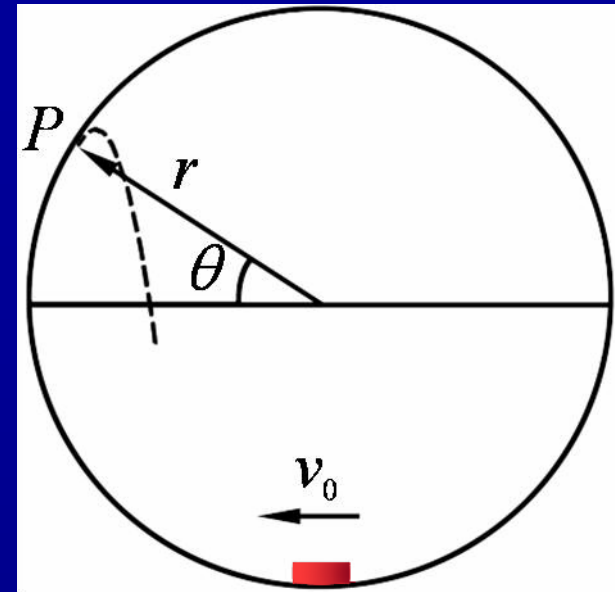
problem3.13 Set $U_{\text{bottom}} = 0$

(a) Conservation of mechanical energy reads

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv_{\text{top}}^2 + 2mgR$$

or

$$\frac{mv_{\text{top}}^2}{R} = \frac{mv_0^2}{R} - 4mg$$



Without losing contact with the track, the block keeps a finite N from the track

$$-N - mg + \frac{mv_{\text{top}}^2}{R} = 0$$

$N \geq 0$ leads

$$-mg + \frac{mv_{\text{top}}^2}{R} \geq 0$$

or

$$-5mg + \frac{mv_0^2}{R} \geq 0$$

$$v_0 \geq \sqrt{5gR} \equiv v_{\text{m}}$$

(b) When $v_0 = 0.775v_m$

at P , conservation law reads

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv_P^2 + mgR(1 + \sin \theta)$$

Normal equation of motion is

$$-N - mg \sin \theta = -\frac{mv_P^2}{R}$$

$N = 0$ leads to

$$\frac{v_P^2}{R} = g \sin \theta$$

or

$$\frac{v_0^2}{R} - 2g(1 + \sin \theta) = g \sin \theta$$

$$\sin \theta = \frac{1}{3} \left(\frac{v_0^2}{gR} - 2 \right) = \frac{1}{3} \left[(0.775)^2 \times 5 - 2 \right] \approx \frac{1}{3}$$

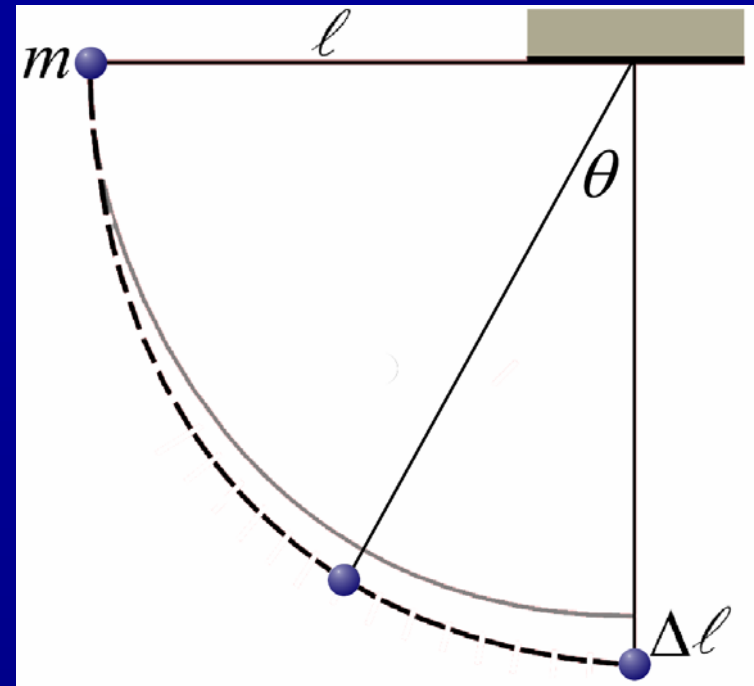
$$\theta = 19^\circ 32'$$

Problem 3.14

$$\begin{aligned} \mathbf{e}_r : \quad & mg \cos \theta - F_T = m(\ddot{r} - r\dot{\theta}^2) \\ \mathbf{e}_\theta : \quad & -mg \sin \theta = m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) \end{aligned}$$

$$F_T' = -k(r - \ell)$$

$$F_T = |F_T'|$$



Initial conditions:

$$t = 0, \theta = \pi/2 \quad r = \ell$$

$$\dot{\theta} = 0 \quad \dot{r} = 0$$

Conservation of energy:

$$\underbrace{\frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - mgr \cos \theta + \frac{1}{2}k(r - \ell)^2}_{r, \theta} = 0 \quad \underbrace{\ell, \pi/2}$$

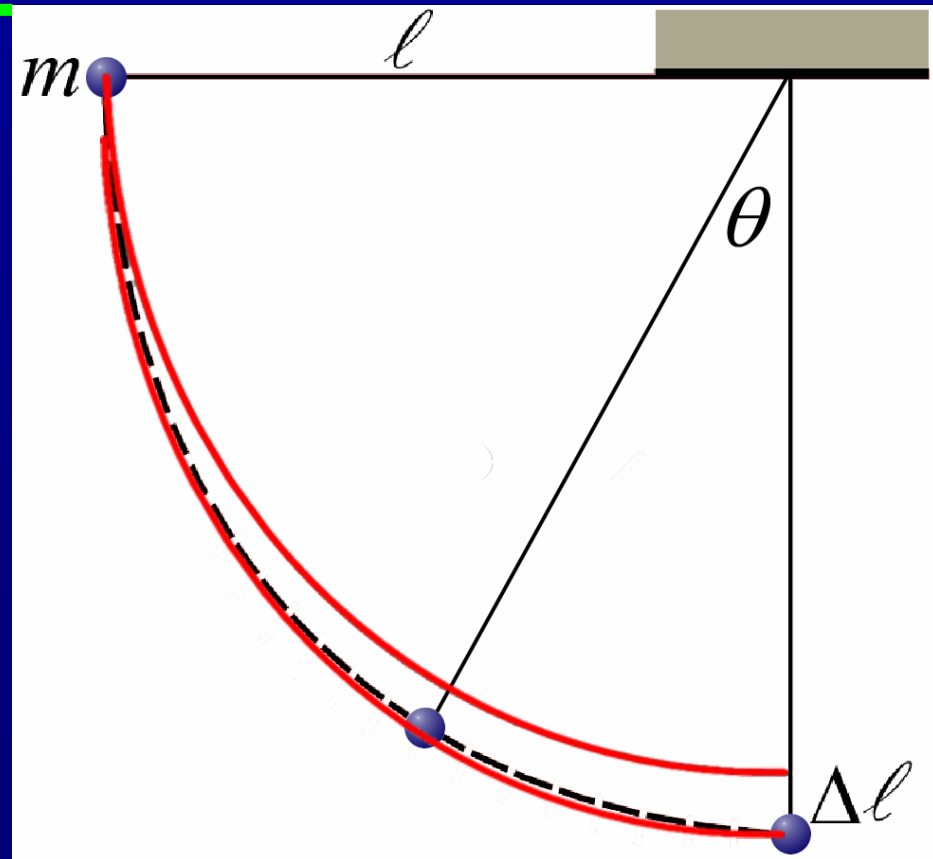
Or for $(\ell + \Delta\ell, 0)$ and $(\ell, \pi/2)$,

$$\frac{1}{2}mv^2 - mg(\ell + \Delta\ell) + \frac{1}{2}k(\Delta\ell)^2 = 0$$

Equation of motion in the normal direction

$$k\Delta\ell - mg = m \frac{v^2}{\ell - \varepsilon\Delta\ell}$$

The radius of
curvature



$$\frac{1}{2}mv^2 - mg(\ell + \Delta\ell) + \frac{1}{2}k(\Delta\ell)^2 = 0 \quad (1)$$

$$mv^2 + mg(\ell - \varepsilon\Delta\ell) - k\Delta\ell(\ell - \varepsilon\Delta\ell) = 0 \quad (2)$$

Eliminating mv^2 terms

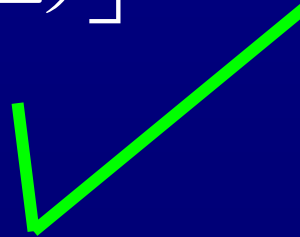
$$-2mg(\ell + \Delta\ell) + k(\Delta\ell)^2 - mg(\ell - \varepsilon\Delta\ell) + k\Delta\ell(\ell + \Delta\ell) = 0$$

$$k\Delta\ell = 3mg \frac{\ell + \frac{2-\varepsilon}{3}\Delta\ell}{\ell + (1-\varepsilon)\Delta\ell}$$

$$\approx 3mg \left(1 + \frac{2\varepsilon - 1}{3} \frac{\Delta\ell}{\ell} \right) \approx 3mg$$

Using Eq.(1)

$$\begin{aligned} v^2 &= 2g(\ell + \Delta\ell) - \frac{k}{m}(\Delta\ell)^2 \\ &= 2g\ell \left(1 + \frac{\Delta\ell}{\ell} - \frac{k\Delta\ell}{2mg} \frac{\Delta\ell}{\ell} \right) \\ &= 2g\ell \left[\underline{\underline{1 + \frac{\Delta\ell}{\ell}}} \left(1 - \frac{k\Delta\ell}{2mg} \right) \right] \\ &= 2g\ell \left(1 - \frac{\Delta\ell}{2\ell} \right) \end{aligned}$$



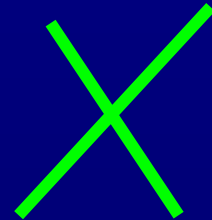
Using (1)+(2)

$$\frac{3}{2}mv^2 = k\Delta\ell(\ell - \varepsilon\Delta\ell) - \frac{1}{2}k(\Delta\ell)^2 + mg(1 + \varepsilon)\Delta\ell$$

$$= k\Delta\ell\ell\left[1 - \left(\frac{1}{2} + \varepsilon\right)\frac{\Delta\ell}{\ell}\right] + mg\ell(1 + \varepsilon)\frac{\Delta\ell}{\ell}$$

$$v^2 = 2g\ell \frac{k\Delta\ell}{3mg} \left[1 - \left(\frac{1}{2} + \varepsilon\right)\frac{\Delta\ell}{\ell}\right] + 2g\ell \frac{1 + \varepsilon}{3} \frac{\Delta\ell}{\ell}$$

$$= 2g\ell \left[1 - \left(\frac{1}{2} + 2\varepsilon\right)\frac{\Delta\ell}{3\ell}\right]$$



Using (2)

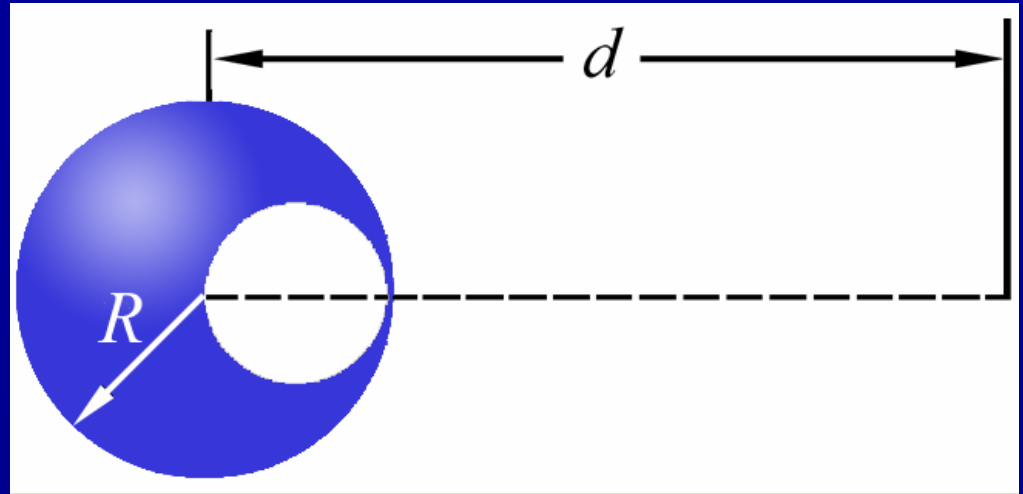
$$mv^2 = (k\Delta\ell - mg)\ell\left(1 - \varepsilon\frac{\Delta\ell}{\ell}\right)$$

$$v^2 = g\ell\left(\frac{k\Delta\ell}{mg} - 1\right)\left(1 - \varepsilon\frac{\Delta\ell}{\ell}\right)$$

$$= 2g\ell\left(1 - \varepsilon\frac{\Delta\ell}{\ell}\right)\left(1 + \frac{2\varepsilon - 1}{2}\frac{\Delta\ell}{\ell}\right)$$

$$= 2g\ell\left(1 - \frac{1}{2}\frac{\Delta\ell}{\ell}\right) \frac{k\Delta\ell}{mg} \rightarrow 3\left(1 + \frac{2\varepsilon - 1}{3}\frac{\Delta\ell}{\ell}\right)$$

Problem 4.4



$$F = G \frac{Mm}{d^2} - G \frac{M \left(\frac{1}{2} \right)^3 m}{(d - R/2)^2}$$

Problem 4.5 Known $h = 640$ km, $m = 220$ kg

a)

$$v_0 = \sqrt{\frac{GM}{R+h}} = 7.54 \times 10^3 \text{ m/s}$$

$$E_0 = -\frac{GMm}{2(R+h)} = -6.25 \times 10^9 \text{ J}$$

b)

$$T_0 = \frac{2\pi(R+h)}{v_0} = 5.84 \times 10^3 \text{ s.}$$

c)

$$E = E_0 - 1.40 \times 10^5 \text{ J} \times 1500 = -6.46 \times 10^9 \text{ J}$$
$$\frac{E}{E_0} = \frac{R+h}{R+r}; \quad r = (R+h) \frac{E}{E_0} - R = 412 \text{ km};$$

$$\frac{v}{v_0} = \sqrt{\frac{R+h}{R+r}} = 1.02$$

$$v = 7.67 \times 10^3 \text{ m/s}$$

$$T = \frac{2\pi(R+h)}{v} = 5.56 \times 10^3 \text{ s}$$

d)

$$F \cdot 2\pi\bar{r} = 1.40 \times 10^5 \text{ J}; \quad F = 3.23 \times 10^{-3} \text{ N}$$

e)

$$\frac{r_0 v_0}{rv} \approx 1.02$$

not conserved

but changes 2% only

Problem 4.6 A person 50kg **can jump ~1m on the earth**, his launch kinetic energy = mgh
This energy is just enough for escape from the planet.

$$\cancel{m}G \frac{m_{\oplus}}{R_{\oplus}^2} h = G \frac{\cancel{m'}\cancel{m}}{R} \longrightarrow \frac{m_{\oplus}}{R_{\oplus}^3} R_{\oplus} h = \frac{m'}{R^3} R^2$$

For **similar density**

$$R \sim \sqrt{R_{\oplus} \cdot 1\text{m}} \approx 2.5\text{km}$$

$$\frac{m_{\oplus}}{R_{\oplus}^3} R_{\oplus} h = \frac{m'}{R^3} R^2$$

For **similar mass**

$$R \sim \frac{R_{\oplus}^2}{1\text{m}} \approx 4 \times 10^7 \text{ km}$$

Which one is reasonable?

Problem 4.7

a) $E = -\frac{GMm}{2r}; \quad E_A + E_B = -\frac{GMm}{r}$

b) $L = 0; \quad v = 0; \quad E = U = -\frac{GM(2m)}{r}$

c) *falls down to the earth*

Problem 4.12 Known $v_a = \sqrt{k/(2ma)}$

$$E_a = \frac{1}{2}mv_a^2 - \frac{k}{a} = -\frac{3k}{4a}$$

$$E_b = \frac{1}{2}mv_b^2 - \frac{k}{b}$$

$$L = av_a = bv_b$$

$$\frac{1}{2}m\left(\frac{a}{b}\right)^2 \frac{k}{2ma} - \frac{k}{b} = -\frac{3k}{4a}$$

$$\frac{1}{4} - \frac{b}{a} = -\frac{3}{4}\left(\frac{b}{a}\right)^2$$

$$3\left(\frac{b}{a}\right)^2 - 4\frac{b}{a} + 1 = 0 \longrightarrow \frac{b}{a} = \frac{1}{6}(4 \pm 2) = \begin{cases} 1 \\ 1/3 \end{cases}$$

$$b = a/3; \quad v_b = 3v_a \quad \text{ellipse}$$

$b = a$ For circle the speed will be

$$v = \sqrt{\frac{k}{ma}} \text{ instead. So it means same point.}$$

$$m \frac{v^2}{a} = \frac{k}{a^2}$$

***Problem 4.14** For central force, we have conservation of angular momentum respect to the origin O . Motion occurs in a plane. In 2D Cartesian system, the equation of motion is

$$-kx = \mu \ddot{x}$$

$$-ky = \mu \ddot{y}$$

The solution is

$$x = A \cos(\omega t + \varphi_1)$$

$$y = B \cos(\omega t + \varphi_2)$$

$$\omega^2 = \frac{k}{\mu}$$

The trajectory is an ellipse. In some initial conditions (say $y_0 = 0, \dot{x}_0 = 0$) it is

$$\left(\frac{x}{x_0}\right)^2 + \frac{k}{\mu} \left(\frac{y}{\dot{y}_0}\right)^2 = 1$$